

BE Literature Survey						
Guide Name:Ms.Pooja B		Div/Batch: BE/B		Title:		Adaptive Hallucination Detection and Reduction Framework for Retrieval-Augmented Large Language Model Chatbots Using AI-Based Response Evaluation
Students Name: 1)Rahul Vishe		2)Sujan Salaskar		Group No:32		
3)Ved Redij		4)Prathamesh Shinde				
Sr.	Paper Title	Author	Publication Details	Summary	Algorithm Used	Research Gaps
1	A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions	Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, Ting Liu	ACM Transactions on Information Systems, Vol. 43, Issue 2, Article 42, pp. 1–55, 2025. DOI: 10.1145/3703155	Comprehensive survey of LLM hallucinations covering definitions, causes, detection, benchmarks and mitigation. It proposes an LLM-specific taxonomy distinguishing factuality hallucination from faithfulness hallucination; faithfulness includes instruction, context and logical inconsistency. It also reviews limitations of RAG for hallucination mitigation.	Survey/taxonomy; reviews factuality and faithfulness detection, benchmarks, mitigation and RAG-based methods.	Does not propose one integrated adaptive detector. It shows hallucination remains unresolved and RAG can still produce unsupported responses, motivating systems that evaluate answers and trigger corrective action.
2	TruthfulQA: Measuring How Models Mimic Human Falsehoods	Stephanie Lin, Jacob Hilton, Owain Evans	Proceedings of ACL 2022, Vol. 1, pp. 3214–3252. DOI: 10.18653/v1/2022.acl-long.229	Introduces TruthfulQA, a benchmark of 817 questions across 38 categories designed to test whether language models reproduce common human misconceptions. The best tested model was truthful on 58% of questions versus 94% for humans, showing that scaling alone does not guarantee truthfulness.	TruthfulQA benchmark; truthfulness evaluation using reference answers and human judgments.	Evaluates general truthfulness rather than whether a response is grounded in retrieved documents. It does not provide RAG-specific detection or adaptive correction.
3	Evaluating the Factual Consistency of Abstractive Text Summarization	Wojciech Kryscinski, Bryan McCann, Caiming Xiong, Richard Socher	Proceedings of EMNLP 2020, pp. 9332–9346. DOI: 10.18653/v1/2020.emnlp-main.750	Introduces FactCC, a weakly supervised model for detecting factual inconsistency between a generated summary and its source. Rule-based corruptions create training data, and the model predicts sentence-level consistency, supporting spans and inconsistent spans.	FactCC; weakly supervised BERT-based factual-consistency classifier with support/conflict span extraction.	Designed for summarization rather than RAG chatbots. It detects factual inconsistency but does not use retrieval-aware evaluation or adaptive regeneration.
4	Asking and Answering Questions to Evaluate the Factual Consistency of Summaries	Alex Wang, Kyunghyun Cho, Mike Lewis	Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL 2020), pp. 5008–5020. DOI: 10.18653/v1/2020.acl-main.450	Introduces QAGS, which generates questions from a model-produced summary, answers them using both the summary and its source, and compares the answers. Mismatches reveal possible factual errors. QAGS showed higher correlation with human factuality judgments than several existing metrics.	QAGS; question generation, question answering and answer-similarity comparison.	Focused on summarization rather than RAG conversations. It evaluates inconsistencies but does not automatically refine prompts or regenerate a detected hallucinated answer.
5	FEQA: A Question Answering Evaluation Framework for Faithfulness Assessment in Abstractive Summarization	Esin Durmus, He He, Mona Diab	Proceedings of ACL 2020, pp. 5055–5070. DOI: 10.18653/v1/2020.acl-main.454	Introduces FEQA, a QA-based metric for measuring faithfulness. Questions are generated from the summary and a QA model answers them using the source document; non-matching answers indicate unsupported information. FEQA correlated better with human faithfulness judgments than several other metrics.	FEQA; question generation + reading-comprehension QA + answer matching.	Designed for summarization and evaluates faithfulness after generation. It does not address RAG-specific retrieval quality or adaptive correction.
6	SummEval: Re-evaluating Summarization Evaluation	Alexander R. Fabbri, Wojciech Kryściński, Bryan McCann, Caiming Xiong, Richard Socher, Dragomir Radev	Transactions of the Association for Computational Linguistics, Vol. 9, pp. 391–409, 2021. DOI: 10.1162/tacl_a_00373	Re-evaluates 14 automatic summarization metrics using outputs from 23 models and expert/crowd human annotations. It also provides a large evaluation toolkit and dataset, highlighting weaknesses and inconsistencies in automatic metrics and the need for better alignment with human judgments.	SummEval benchmark/evaluation toolkit; comparison of automatic metrics against human judgments.	Focuses on summarization rather than RAG and does not provide hallucination-specific adaptive correction. It supports the need to validate automated evaluators against human judgments.
7	HalluLens: LLM Hallucination Benchmark	Yejin Bang, Ziwei Ji, Alan Schelten, Anthony Hartshorn, Tara Fowler, Cheng Zhang, Nicola Cancedda, Pascale Fung	Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025), Vol. 1, pp. 24128–24156. DOI: 10.18653/v1/2025.acl-long.1176	Introduces HalluLens, a benchmark using a taxonomy that distinguishes intrinsic and extrinsic hallucinations. It emphasizes extrinsic hallucination and introduces three new extrinsic evaluation tasks with dynamic test-set generation to reduce data leakage and improve robustness.	HalluLens benchmark; intrinsic/extrinsic hallucination taxonomy and dynamic test-set generation.	Provides a benchmark rather than an end-to-end correction system. It is not specifically designed for document-grounded RAG responses or adaptive regeneration.
8	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks	Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, Douwe Kiela	Advances in Neural Information Processing Systems 33 (NeurIPS 2020).	Introduces RAG, combining a pretrained sequence-to-sequence generator with a neural retriever over a dense non-parametric knowledge index. It presents RAG-Sequence and RAG-Token and reports state-of-the-art results on three open-domain QA tasks, with more specific, diverse and factual generation than a parametric-only baseline.	RAG; neural retriever, dense Wikipedia index, seq2seq generator; RAG-Sequence and RAG-Token.	RAG improves external knowledge access but depends on retrieval quality and does not verify whether the generated answer is faithful to retrieved evidence. It has no hallucination detection or adaptive regeneration.
9	Dense Passage Retrieval for Open-Domain Question Answering	Vladimir Karpukhin, Barlas Öğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, Wen-tau Yih	Proceedings of EMNLP 2020, pp. 6769–6781. DOI: 10.18653/v1/2020.emnlp-main.550	Introduces DPR, replacing traditional sparse retrieval with learned dense representations. Separate BERT-based question and passage encoders map queries and passages to vectors, and relevant passages are retrieved using vector similarity. DPR improved top-20 retrieval accuracy over BM25 by 9–19 percentage points across evaluated datasets.	DPR; BERT dual encoder, dense embeddings and dot-product similarity.	Improves retrieval but does not address answer generation, hallucination detection or response verification. Retrieval errors can still propagate to generated answers.

BE Literature Survey						
Guide Name:Ms.Pooja B			Div/Batch: BE/B		Title:	Adaptive Hallucination Detection and Reduction Framework for Retrieval-Augmented Large Language Model Chatbots Using AI-Based Response Evaluation
Students Name: 1)Rahul Vishe		2)Sujan Salaskar		Group No:32		
3)Ved Redij		4)Prathamesh Shinde				
Sr.	Paper Title	Author	Publication Details	Summary	Algorithm Used	Research Gaps
10	Retrieval Augmented Language Model Pre-Training	Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, Ming-Wei Chang	Proceedings of ICML 2020, PMLR 119, pp. 3929–3938.	Introduces REALM, integrating a latent knowledge retriever into language-model pre-training. The retriever accesses a large corpus such as Wikipedia during pre-training, fine-tuning and inference. REALM demonstrated 4–16 percentage-point absolute gains on three open-domain QA benchmarks.	REALM; latent neural knowledge retriever, unsupervised retrieval pre-training and masked language-modeling objective.	Focuses on retrieval and pre-training rather than post-generation hallucination detection. It does not verify final answers against retrieved evidence or adaptively correct them.
11	Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering	Gautier Izacard, Edouard Grave	Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2021), pp. 874–880. DOI: 10.18653/v1/2021.eacl-main.74	Studies how generative QA models benefit from retrieved evidence and introduces Fusion-in-Decoder (FiD). Retrieved passages are encoded independently and their representations are combined by the decoder. The method achieved strong results on Natural Questions and TriviaQA and benefited from increasing the number of retrieved passages.	Fusion-in-Decoder (FiD); passage retrieval, independent passage encoding and decoder-level evidence fusion.	Improves multi-passage evidence aggregation but does not verify factual faithfulness or provide adaptive hallucination correction.
12	Atlas: Few-shot Learning with Retrieval Augmented Language Models	Gautier Izacard, Patrick Lewis, Maria Lomeli, Lucas Hosseini, Fabio Petroni, Timo Schick, Jane Dwivedi-Yu, Armand Joulin, Sebastian Riedel, Edouard Grave	Journal of Machine Learning Research, Vol. 24, No. 251, pp. 1–43, 2023.	Introduces Atlas, a retrieval-augmented language model designed for knowledge-intensive tasks with very few training examples. It evaluates MMLU, KILT and Natural Questions and studies updateable document indexes. Atlas achieved over 42% accuracy on Natural Questions using only 64 examples.	Atlas; retrieval-augmented language model with jointly trained retriever and encoder-decoder generator.	Demonstrates few-shot RAG performance but does not focus on post-generation hallucination detection, evaluator-driven correction or adaptive refinement.
13	RAGBench: Explainable Benchmark for Retrieval-Augmented Generation Systems	Robert Friel, Masha Belyi, Atindriyo Sanyal	arXiv preprint arXiv:2407.11005, 2024.	Introduces RAGBench, a 100,000-example benchmark covering five industry-specific domains and multiple RAG task types. It also proposes TRACe, an explainable and actionable evaluation framework for diagnosing RAG systems. The study finds that LLM-based RAG evaluation can struggle to compete with a fine-tuned RoBERTa model.	RAGBench dataset and TRACe evaluation framework; supervised and LLM-based RAG evaluation approaches.	Provides comprehensive evaluation and actionable feedback but does not establish a complete adaptive loop that automatically uses evaluation results to regenerate and re-evaluate responses.
14	Benchmarking Retrieval Strategies for Biomedical Retrieval-Augmented Generation: A Controlled Empirical Study	Devi Prasad Bal, Subhashree Puhan	arXiv preprint arXiv:2605.02520, submitted May 4, 2026. DOI for archived version: 10.5281/zenodo.19774692.	Presents a controlled comparison of five retrieval strategies—Dense Vector Search, Hybrid BM25 + Dense retrieval, Cross-Encoder Reranking, Multi-Query Expansion and Maximal Marginal Relevance (MMR)—in a biomedical QA RAG pipeline. All strategies use the same GPT-4o-mini generator, ChromaDB and text-embedding-3-small. On 250 BioASQ-derived question-answer pairs, cross-encoder reranking achieved the best composite score (0.827) and contextual precision	Dense retrieval, hybrid BM25+dense retrieval, cross-encoder reranking, multi-query expansion and MMR; evaluated with DeepEval.	Focuses on retrieval-strategy comparison in a biomedical domain rather than hallucination detection or adaptive correction. It does not study a post-generation feedback/regeneration loop.
15	G-EVAL: NLG Evaluation using GPT-4 with Better Human Alignment	Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, Chenguang Zhu	Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023), pp. 2511–2522.	Introduces G-Eval, an LLM-based evaluation framework using chain-of-thought reasoning and a form-filling paradigm. The evaluator receives task instructions and criteria, generates evaluation steps and scores the output. GPT-4 G-Eval achieved a Spearman correlation of 0.514 with human judgments on summarization.	G-Eval; GPT-4/LLM-as-judge, chain-of-thought evaluation steps and form-filling scoring.	LLM evaluators can exhibit bias, including preference toward LLM-generated text. G-Eval does not itself provide RAG-specific evidence verification or adaptive regeneration.
16	Self-Refine: Iterative Refinement with Self-Feedback	Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, Peter Clark	Advances in Neural Information Processing Systems 36 (NeurIPS 2023). arXiv:2303.17651.	Proposes an iterative approach where an LLM generates an initial response, provides feedback on its own output and refines it repeatedly. The same LLM acts as generator, feedback provider and refiner, without additional supervised training or reinforcement learning. Across seven tasks, performance improved by about 20% absolute on average.	Self-Refine; generate → self-feedback → refinement loop using one LLM.	Feedback is not necessarily grounded in trusted external evidence and the method is not specifically designed for RAG hallucination detection. Evidence-grounded evaluation before refinement remains an open direction.
17	Chain-of-Verification Reduces Hallucination in Large Language Models	Shehzaad Dhuliawala, Mojtaba Komeili, Jing Xu, Roberta Raileanu, Xian Li, Asli Celikyilmaz, Jason Weston	Findings of ACL 2024. DOI: 10.18653/v1/2024.findings-acl.212	Introduces Chain-of-Verification (CoVe) to reduce hallucinations by separating answer generation from verification. The model drafts an answer, plans verification questions, answers those questions independently, and generates a final verified response. Experiments on list-based questions, MultiSpanQA and long-form generation show reduced hallucination.	Chain-of-Verification; draft → verification-question planning → independent verification → final answer.	Not specifically designed around retrieved RAG context or multi-dimensional RAG evaluation. It does not establish a general retrieval-aware adaptive loop for domain-specific knowledge bases.
18	Fine-Refine: Iterative Fine-grained Refinement for Mitigating Dialogue Hallucination	Xiangyan Chen, Yujian Gan, Matthew Purver	arXiv preprint arXiv:2602.15509, submitted February 17, 2026.	Introduces Fine-Refine, which decomposes a response into atomic units, verifies each unit using external knowledge, evaluates fluency with perplexity and iteratively corrects granular errors. On HybriDialogue and OpendialKG, it achieved up to a 7.63-point improvement in dialogue fact score with a small trade-off in dialogue quality.	Fine-Refine; atomic claim decomposition, external verification, perplexity-based fluency assessment and iterative refinement.	Targets dialogue hallucination rather than a RAG knowledge-base chatbot. It does not provide a general framework combining RAG retrieval, multi-dimensional evaluation and adaptive regeneration.

BE Literature Survey						
Guide Name:Ms.Pooja B		Div/Batch: BE/B		Title:		Adaptive Hallucination Detection and Reduction Framework for Retrieval-Augmented Large Language Model Chatbots Using AI-Based Response Evaluation
Students Name: 1)Rahul Vishe		2)Sujan Salaskar		Group No:32		
3)Ved Redij		4)Prathamesh Shinde				
Sr.	Paper Title	Author	Publication Details	Summary	Algorithm Used	Research Gaps
19	Legal Query RAG	Rahman S. M. Wahidur, Sumin Kim, Haeung Choi, David S. Bhatti, Heung-No Lee	IEEE Access, Vol. 13, pp. 36978–36994, 2025. DOI: 10.1109/ACCESS.2025.3542125.	Introduces LQ-RAG, a legal-domain RAG framework with recursive feedback. It combines an evaluation agent, a prompt-engineering agent, a fine-tuned legal embedding model and a specialized response generator. The evaluation agent checks response quality; if criteria are not met, feedback is passed to the prompt-engineering agent for another retrieval/generation cycle.	LQ-RAG; recursive feedback, audit agent, prompt-engineering agent, fine-tuned legal embeddings and specialized generator.	This is one of the closest prior works to an evaluation-plus-regeneration loop, so that combination should not be claimed as completely novel. Its main limitation for our project is its legal-domain specialization; a broader domain-independent framework with systematic multi-dimensional evaluation and validation would be a clearer research direction.
20	Towards Trustworthy Retrieval Augmented Generation for Large Language Models: A Survey	Bo Ni, Zheyuan Liu, Leyao Wang, Yongjia Lei, Yuying Zhao, Xueqi Cheng, Qingkai Zeng, Luna Dong, Yinglong Xia, Krishnaram Kenthapadi, Ryan Rossi, Franck Dernoncourt, Md Mehrab Tanjim, Nesreen Ahmed, Xiaorui Liu, Wenqi Fan, Erik Blasch, Yu Wang, Meng Jiang, Tyler Derr	arXiv preprint arXiv:2502.06872, 2025.	Surveys trustworthy RAG and explains that, although RAG can improve reliability and reduce hallucination, it also introduces risks involving robustness, privacy, adversarial attacks, fairness, explainability and accountability. The survey organizes the area around reliability, privacy, safety, fairness, explainability and accountability and identifies future research directions.	Survey/taxonomy of trustworthy RAG across reliability, privacy, safety, fairness, explainability and accountability.	Does not implement a specific adaptive hallucination detector or correction loop. It identifies broad trustworthiness challenges, leaving room for a practical framework focused on evidence-grounded hallucination detection, multi-dimensional response evaluation and adaptive reduction.